

Hearing and Tinnitus

Hearing loss of age (KLHDC7B, CTBP2, ILDR1, TRIOBP, EYA4, LMX1A, ISG20/ACAN),
connexin 26 (GJB2), ototoxicity, tinnitus, vertigo: polygenic scores and loci load

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is
neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The hearing loss of age is typical in the set: auditory difficulty in percentile 46.1, hearing loss in the 62.1 and sensorineural loss in the 58.3, with loci load in the 54.2. There are no risk alleles of **KLHDC7B**, **CTBP2** or **LMX1A**. The 26 (GJB2) connexin is normal (without M34T or V37I) and the **ACYP2** does not have the cisplatin ototoxicity allele. The vertigo and Ménière's disease are very low in the loci (0.2).

Unfavourable. Tinnitus is the weak point. The genomic score is in percentile 89.7, the severity of tinnitus in 94.6 and the loci load in 99.8; only the small score of 18 thousand variants is in 24.6 (\$46). The use of hearing aids is in percentile 90.0 (\$46). It has 2 risk alleles in **ISG20/ACAN** and **EYA4** and 1 in **ILDR1**.

What does it mean? The Sérgio F ear has a typical genetic cochlea, but a central auditory pathway with a tendency to tinnitus. Tinnitus is often the first sign of noise injury, so protecting the ears from noise is the measure with the most return. Untreated hearing loss is one of the major modifiable factors of cognitive decline. Tadalafil (class of PDE5 inhibitors) has a rare warning of sudden hearing loss, and isotretinoin rarely causes tinnitus: if they appear, they should be communicated soon.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 10 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 5 of the chip, 4 imputed and 1 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 7 PGS Catalog scores calculated locally with 2.960.263 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 234 loci, 175 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a high percentile = more risk (auditory difficulty, hearing aid, hearing loss, tinnitus). Four of them (auditory difficulty, hearing aid, tinnitus of 18 thousand variants and severity of tinnitus) are those of §46, with the same values.

Hearing loss

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hearing difficulties	PGS000762 (Cherny 2020, 100 thousand variants)	86.215 100.325 (weighted coverage 87%)	46,1	TIPIC	typical
Use of hearing aids	PGS000763 (Cherny 2020, 4270 variants)	3.847 4.270 (weighted coverage 90%)	90,0	MAIS PROVABLE	unfavourable
Hearing loss	PGS019114 (Chen 2025, PRS-CS, 984 thousand variants)	950.712 983.822 (weighted coverage 96%)	62,1	TIPIC	typical
Sensory hearing loss	PGS019119 (Chen 2025, PRS-CS, 984 thousand variants)	950.711 983.819 (weighted coverage 95%)	58,3	TIPIC	typical

- **Hearing difficulties:** Já no §46.
- **Use of hearing aids:** Já no §46.

Tinnitus

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Tinnitus	PGS019124 (Chen 2025, PRS-CS, 984 thousand variants)	950.707 983.817 (weighted coverage 95%)	89,7	MAIS PROVABLE	unfavourable
Tinnitus	PGS019123 (Chen 2025, MegaPRS, 18 thousand variants)	17.786 18.193 (weighted coverage 98%)	24,6	TIPIC	typical
Tinnitus severity	PGS001533 (Tanigawa 2022, 343 variants)	285 343 (weighted coverage 84%)	94,6	MAIS PROVABLE	unfavourable

- **Tinnitus:** Já no §46.
- **Tinnitus severity:** Já no §46.

4. Key variants by topic

Replicated loci of hearing loss of age and variants of monogenic deafness and ototoxicity.

Hearing loss of age

Genes of hair cells, synapses and inner ear structures identified in the GWAS of hearing loss of age. Each allele changes little the risk.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
KLHDC7B	rs36062310		G/G	imputation DR2=0.90	A (0 copy/s)	favourable	Hearing loss of age (OR ~1.15): outer hair cells	Moderate
CTBP2	rs10901863		C/C	imputation DR2=0.80	T (0 copy/s)	favourable	Hearing loss of age (synapsis in internal hair cell)	Moderate
ILDR1	rs2877561		A/C	chip	A (1 copy/s)	unfavourable (light)	Hearing loss of age (OR ~1.14; tricellular junctions of the organ of Corti)	Moderate
ISG20/ACAN	rs12594617		C/C	imputation DR2=1.00	C (2 copy/s)	unfavourable	Head locus of sensorineural hearing loss in adults	Moderate
TRIOBP	rs58389158		SNP not found in the data	SNP not found in the data	T faux copy/s)	without data	Hearing difficulties in adults (stereocilia of hair cells)	Moderate
EYA4	rs9493627		A/A	chip	A (2 copy/s)	unfavourable	Hearing loss of age (G277S; DFNA10 deafness gene)	Weak
LMX1A	rs7525101		C/C	imputation DR2=0.99	T (0 copy/s)	favourable	Hearing loss of age (development of the inner ear)	Weak

Monogenic deafness and ototoxicity

Connexin 26 (GJB2) is the most common genetic cause of recessive deafness. ACYP2 is associated with hearing loss caused by cisplatin.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GJB2 (connexin 26)	rs35887622	M34T	A/A	chip	G (0 copy/s)	favourable	M34T: mild recessive hearing loss variant (low penetration)	Moderate
GJB2 (connexin 26)	rs72474224	V37I	C/C	chip	T (0 copy/s)	favourable	V37I: mild to moderate recessive hearing loss variant	Moderate
ACYP2	rs1872328		G/G	chip	A (0 copy/s)	favourable	Allele A is associated with cisplatin ototoxicity (relevant only if there is chemotherapy with platins)	Moderate

5. GWAS loci burden

Sum of the loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of hearing loss of age (auditory difficulty, hearing aid, sensorineural loss), tinnitus and vertigo / Ménière's disease. Alleles are oriented towards more risk and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Hearing loss of age	142 (115)	100 de 230	115	54,2	typical
Tinnitus	54 (44)	58 de 88	44	99,8	unfavourable
Vertigo and Ménière's disease	38 (16)	5 de 32	16	0,2	favourable

6. Favourable

- **Hearing loss of typical age:** hearing impairment in percentile 46.1, hearing loss in the 62.1, sensorineural in the 58.3; loci load in the 54.2. **Moderate.**
- **KLHDC7B G/G, CTBP2 C/C and LMX1A C/C:** without risk alleles. **Moderate.**
- **GJB2 without M34T or V37I** (A/A e C/C). **Moderate.**
- **ACYP2 rs1872328 G/G:** without cisplatin ototoxicity allele. **Moderate.**
- **Vertigo and Ménière's disease:** loci load in 0.2. **Weak moderated.**

7. Unfavourable

- **Tinnitus** in percentile 89.7 (genomic score), gravity in 94.6 and loci load in 99.8 (the small score of \$46 is in 24.6). **Moderate.**
- **Use of hearing aids** in percentile 90.0 (\$46). **Moderate.**
- **ISG20/ACAN rs12594617 C/C** (2 alleles; leading locus of sensorineural loss), **EYA4 rs9493627 A/A** (2 alleles) and **ILDR1 A/C** (1 allele). **Weak moderated.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Oxidative stress	Noise and ototoxic drugs damage free radical hair cells	Conditional
Mitochondrial dysfunction	The ciliated cochlea cells depend heavily on mitochondria (aminoglycosides and mitochondrial variants)	Neutral
Vascular Aging	Vascular stria and cochlea microcirculation age with glucose, cholesterol and tobacco	Conditional
Neuroinflammation	Tinnitus involves hyperactivity and reorganization of the central auditory pathway; tinnitus in percentile 89.7	Unfavourable
Exhaustion of Staminary Cells	The hair cells do not regenerate in mammals; the loss accumulates throughout life	Neutral
Sleep disturbance	Tinnitus worsens with short sleep and stress, and disturbs falling asleep	Unfavourable (conditional)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Protecting the ears from noise:** protectors (foam or filter plugs for music) in concerts, discos, works, electric tools and motorized sports. Rule 60/60 in the headphones (maximum 60% of the volume during 60 minutes in a row) and headphones with noise cancellation instead of increasing the volume. *Suggestion (not applied).*
2. **Base audiogram** (including high frequencies) and repeat each 3–5 years, or earlier if persistent tinnitus appears or difficulty in hearing with background noise. *Suggestion (not applied).*
3. **Sudden hearing loss is an urgency:** if it happens (especially in an ear), stop tadalafil and go to the emergency of otorhinolaryngology in the first 72 hours (corticoids work better early). *Suggestion (not applied).*
4. **Ototoxic drugs:** to remind the physician about the risk with aminoglycosides (gentamicin), high-dose anxiolytics, high-dose salicylates and cisplatin. ACYP2 is favourable and the mitochondrial profile did not show pathogenic variants, but high genetic tinnitus calls for caution. *Suggestion (not applied).*
5. **Isotretinoin:** if tinnitus or hearing disorders arise, communicate to the dermatologist. *Suggestion (not applied).*
6. **If tinnitus appears:** sound therapy (sound at night), cognitive-behavioral therapy for tinnitus, treating associated hearing loss and protecting sleep (melatonin 0.3–1 mg of the plane). Caffeine $\leq$ 200 mg only in the morning and alcohol zero (already in the plane). *Suggestion (not applied).*
7. **If hearing loss arises:** early hearing aid, without waiting. It reduces social isolation and the risk of cognitive decline. *Suggestion (not applied).*
8. **Cochlea vascular health:** glucose, low cholesterol (LDL/ApoB, already in the plane), aerobic exercise and zero tobacco protect the inner ear microcirculation. *Suggestion (not applied).*

10. Limitations

- The chip reads few variants of monogenic deafness; the 35delG deletion of GJB2 comes from imputation (§77) and mitochondrial variants (m.1555A>G) depend on the quality of mitochondrial probes.
- The tinnitus scores disagree (the small is typical, the genomic and gravity are high); the loci load supports the high.
- Tinnitus and hearing loss depend heavily on exposure to noise throughout life, which the genes do not measure.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (7 scores: 4 of hearing loss and 3 of tinnitus; 4 already used in §46). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510-8$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** Wells 2019 and Ivarsdottir 2021 (age loss), Trpchevska 2022 and Clifford 2024 (zombido), Cherny 2020 (scores), Xu 2015 (ACYP2 and cisplatin), Livingston 2024 (hearing and dementia), regulatory alerts on PDE5 inhibitors and sudden hearing loss, GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Hearing loss of age	KLHDC7B	rs36062310	G/G	imputation DR2=0.90	A	0	strand +	without the risk allele
Hearing loss of age	CTBP2	rs10901863	C/C	imputation DR2=0.80	T	0	strand +	without the risk allele
Hearing loss of age	ILDR1	rs2877561	A/C	chip	A	1	strand +	1 copy of risk allele
Hearing loss of age	ISG20/ACAN	rs12594617	C/C	imputation DR2=1.00	C	2	strand + (palindromic; assumed +)	2 risk allele copies
Hearing loss of age	TRIOBP	rs58389158	SNP not found in the data	SNP not found in the data	T			SNP not found in the data
Hearing loss of age	EYA4	rs9493627	A/A	chip	A	2	strand +	2 risk allele copies
Hearing loss of age	LMX1A	rs7525101	C/C	imputation DR2=0.99	T	0	strand +	without the risk allele
Monogenic deafness and ototoxicity	GJB2 (connectin 26)	rs35887622	A/A	chip	G	0	strand +	without the risk allele
Monogenic deafness and ototoxicity	GJB2 (connectin 26)	rs72474224	C/C	chip	T	0	strand +	without the risk allele
Monogenic deafness and ototoxicity	ACYP2	rs1872328	G/G	chip	A	0	strand +	without the risk allele